

ONTARIO, CA, USA

WMO: 747040

Lat: 34.056N

Lon: 117.600W

Elev: 949

StdP: 14.20

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 03102

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	38.7	41.0	3.5	6.8	66.1	9.6	9.3	64.3	33.5	62.4	28.0	62.4	2.8	40	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	27.5	100.4	69.2	97.6	68.3	94.8	67.4	73.0	93.6	71.4	91.4	70.1	89.4	12.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	100.2	79.8	64.4	93.8	77.7	63.5	90.7	76.0	37.0	93.5	35.8	91.7	34.5	89.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	17.4	15.5	DB	32.7	107.5	3.5	3.3	30.2	109.9	28.1	111.8	26.1	113.7	23.5	116.0	(4)
(5)			WB	28.6	75.6	2.8	2.3	26.6	77.3	25.0	78.6	23.4	79.9	21.4	81.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.5	56.4	56.4	60.3	63.2	67.1	72.6	78.7	79.7	76.7	69.2	61.6	55.4	(6)	
	DBStd	10.39	6.04	6.00	6.22	6.62	6.13	5.73	4.75	4.97	6.27	6.41	6.38	5.43	(7)	
	HDD50	38	11	9	2	1	0	0	0	0	0	0	2	13	(8)	
	HDD65	1323	273	247	173	110	46	7	0	0	1	29	138	299	(9)	
	CDD50	6065	210	189	320	398	529	679	890	921	801	595	351	182	(10)	
	CDD65	1878	7	8	26	57	110	236	425	456	353	160	37	3	(11)	
	CDH74	20211	129	167	413	755	1141	2350	4450	4938	3630	1654	509	75	(12)	
	CDH80	10170	16	36	122	286	486	1126	2412	2790	1964	755	171	6	(13)	
(14) Wind	WSAvg	6.0	5.1	5.2	6.0	6.8	7.1	7.0	6.9	6.4	5.8	5.6	4.7	5.0	(14)	
(15) Precipitation	PrecAvg	22.4	5.0	5.6	3.0	1.5	0.6	0.2	0.1	0.1	0.2	0.9	1.4	3.6	(15)	
	PrecMax	39.9	20.9	16.3	11.5	4.4	5.6	1.3	0.6	0.4	1.2	6.4	3.2	11.9	(16)	
	PrecMin	9.5	0.4	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.6	(17)	
	PrecStd	9.2	5.4	4.1	3.0	1.1	1.2	0.4	0.2	0.1	0.3	1.4	0.9	2.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.2	84.6	89.4	93.6	97.0	101.5	103.2	104.6	103.9	97.9	90.5	80.8	(19)	
		MCWB	54.8	56.4	58.8	60.1	64.4	68.4	70.7	70.6	68.1	62.9	58.5	53.7	(20)	
	2%	DB	77.8	79.4	83.4	87.6	90.5	95.3	98.6	100.1	99.3	92.9	85.3	75.5	(21)	
		MCWB	53.0	54.6	57.8	59.1	63.0	67.6	70.9	69.9	68.6	61.7	57.4	52.3	(22)	
	5%	DB	73.6	74.1	78.7	82.3	85.6	90.5	95.2	97.1	95.4	88.5	80.0	71.8	(23)	
		MCWB	51.9	53.2	56.5	58.1	62.5	66.5	69.3	68.7	67.3	60.5	55.8	51.0	(24)	
	10%	DB	69.7	68.6	73.4	77.6	80.5	86.8	92.1	93.4	91.0	83.5	74.9	67.1	(25)	
		MCWB	50.8	52.1	55.4	56.7	61.9	65.9	68.6	68.1	66.0	59.6	54.0	50.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	60.6	63.9	65.2	69.8	72.2	75.6	75.9	73.3	69.4	62.4	59.4	(27)	
		MCDB	69.6	74.7	79.0	85.1	89.5	91.9	97.0	95.6	95.5	84.0	73.6	66.4	(28)	
	2%	WB	57.2	57.9	61.0	62.7	66.8	69.7	73.0	73.0	71.3	66.5	60.3	57.3	(29)	
		MCDB	66.0	68.1	76.2	80.4	84.7	89.6	93.2	93.7	91.8	81.1	73.3	65.4	(30)	
	5%	WB	55.4	56.1	59.1	60.3	64.6	68.2	71.3	71.2	69.7	64.6	58.8	55.4	(31)	
		MCDB	63.9	67.1	72.1	75.7	80.7	86.6	90.9	91.4	88.9	79.3	72.2	62.8	(32)	
	10%	WB	53.9	54.5	57.3	58.6	62.7	66.6	69.8	69.5	68.2	62.8	57.3	53.7	(33)	
		MCDB	63.2	64.6	69.1	72.7	76.6	83.4	88.1	89.4	85.5	76.3	69.4	61.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.8	19.8	21.5	22.3	21.9	24.9	26.8	27.5	25.8	23.5	23.0	20.8	(35)	
		MCDBR	28.7	30.1	30.5	31.2	30.9	31.1	29.5	30.4	30.8	31.9	31.3	29.2	(36)	
	5% WB	MCWBR	12.9	12.5	12.4	12.0	11.3	10.3	9.9	10.5	10.3	11.5	12.5	13.3	(37)	
		MCWBR	20.3	22.4	25.2	26.8	27.6	28.9	27.6	27.9	27.4	26.3	24.8	19.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.313	0.335	0.350	0.352	0.362	0.377	0.361	0.345	0.338	0.301	0.300	(40)	
		taud	2.544	2.534	2.448	2.413	2.437	2.422	2.442	2.474	2.505	2.496	2.591	2.587	(41)	
	Ebn at Noon		290	299	298	297	295	291	286	290	291	285	289	285	(42)	
		Edh at Noon	26	29	34	37	36	37	36	34	32	30	24	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		1022	1307	1759	2141	2348	2523	2430	2299	1957	1490	1139	927	(44)	
	RadStd		71	128	88	96	124	128	115	61	59	90	55	64	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.10	N/A	-5.17	N/A	N/A	N/A	N/A	-148	+425	+248
(47) Regional (51 neighbors)		+1.13	N/A	-3.06	+1.30	N/A	N/A	N/A	-202	+401	+202

Nomenclature: See separate page